

A Multi-Model Approach to Forecasting the Olympics Medal Counts

Interpretability vs. Accuracy

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Abstract

Forecasting Olympic success requires untangling a complex web of economic, demographic, and historical variables. This study cuts through that complexity to predict medal counts for the upcoming Summer Games by analyzing key determinants, including GDP, historical performance, Olympic team size and social indicators such as the Human Development Index (HDI) and Gender Development Index (GDI). The project employs a dual-modeling approach to address the trade-off between theoretical interpretability and predictive accuracy. First, a machine learning method (Random Forest) is used to maximize precision and accuracy through non-linear feature capture. Second, a complementary statistical model (Zero-Inflated Negative Binomial) is implemented to ensure the interpretability of results. By combining those distinct methodologies, the research provides both a valid prediction for the next Olympics and also a deeper understanding of the different factors behind national athletic success.

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Chapter 1

Introduction

The Olympics offer an interesting data science challenge. It is not just about multiple sports with as many datasets and numbers associated; it is a reflection of national prosperity, institutional wealth and geopolitical shifts. The Olympics now are a multi-billion dollar business. Costs have been growing for years, often exceeding $10 billion per event; for Rio 2016, that figure hit $23.6 billion (Budzier and Flyvbjerg (2024)), against roughly $3 to $5 billion in generated revenue (Zimbalist (2017)). As the business grows, so does the demand for accurate forecasting (for nations, sponsors, organizers...).

Moreover, these forecasts are also used by the national Olympic committees to justify multi-million dollar budgets for athlete development and to transition from a "gut feeling" scouting to a high stakes macro level model. Indeed, nations use the athletic success to enhance their international prestige and foster diplomatic ties, for instance the spread of diplomatic channels between United States and China in the 1970s (Qingmin (2013)). Beyond those global goals, sporting excellence cultivates a sense of national identity, pride and promotes public health (Nieto et al. (2024)).

From a statistical perspective, the Olympics are a significantly harder data problem than traditional sports like football or basketball. This is due to the fact that the quadrennial cycle yields a sample size that lacks the statistical power required for robust causal inference, particularly when compared to the dense datasets of annual macro-variables like GDP. Besides this, the "Host Effect" is also an issue that has to be taken into consideration. Hosting is not just about "home-field" advantage, but also about massive infrastructure investment and automatic qualification slots. In a data science analysis, this effect has to be weighted in order to have a complete framework. Finally, geopolitical developments also play a critical role in this context. For this reason, this study examines historical dissolutions, such as those of the Soviet Union and Yugoslavia, as well as more recent conflicts in Ukraine and the Middle East, all of which are

factored into the final analysis.

The literature about understanding and explaining Olympic data and results has as one of its pillars the article of Bernard and Busse (2004). The authors examine the determinants of Olympic success at the country level; for instance, whether a specific nation wins its fair share of Olympic medals, taking into account the role of population and economic resources.

The literature of the most recent years has seen an increasing number of articles about medal prediction, highlighting the importance of a problem that sits at the intersection of athletic performance, geopolitics, and macroeconomics.

Numerous techniques have been created and enhanced to suit this problem in the most precise way. However, methods like XGBoost (Li et al. (2025)), or HEAH Framework (Jin (2025)), have prioritized precision over transparency and interpretability, with the goal of achieving the highest accuracy possible. The term interpretability on the one hand describes how straightforward it is to a human to comprehend the technical details of a model, without getting involved in mathematical jargon. On the other hand, accuracy also needs to be taken into account; it measures how close the results of the model are compared to the real world (Shukla and Tripathi (2012)). Those two concepts are a trade-off and an increase in either may cause a decrease in the other.

1.1 Focus on Social-Economic Indicators

However, the articles about medal prediction mentioned so far do not include social-economic indicators such as Gender Development Index (GDI) and Human Development Index (HDI).

The GDI accounts for gender inequalities in three basic dimensions of human development: health, measured by expectancy at birth; education, measured by both the expected years of schooling and the mean years of schooling for adults ages 25 years and older; finally, command over economic resources and power, measured by female and male estimated earned salary (United Nations Development Programme (2024a)).

The HDI, instead, takes into account the average achievement in key dimensions of human development: for instance, living a healthy life, being knowledgeable and having a decent standard of living. The HDI takes those three indicators and calculates their geometric mean (United Nations Development Programme (2024b)).

Even if GDP is in fact able to measure the raw economical power of a nation, it may fail in measuring how efficiently a country converts that power into athletic excellence. Alternatively including indicators like GDI and HDI might be able to expand the explanations in a deeper way; for example, GDI captures the fact that if a country ignores 50% of its talent pool (women) due to social inequality, its "effective" population for the Olympics is halved. Instead HDI pulls in health and education, which are the true foundations of an athlete's development.

However, those two variables are not statistically independent, given the fact that they are derived from similar features, as a consequence it is important to take this possible multicollinearity into account (Schüler (2006)).

1.2 Modeling Technique

As discussed before, ensemble methods (Random Forest, XGBoost) in sports analytics are gaining popularity due to their accuracy scores, but, alongside the "What", every analysis needs the "Why" as well, i.e. a model that allows us to see how much interpretability we lose when we chase maximum accuracy. For this reason in this analysis, two models will be juxtaposed: one on side the Random Forest with the goal of maximum accuracy for the medals' predictions. Achieving a high R^2 (as well as a low *Mean Absolute Error* MAE) and a number of predicted medals as similar to the actual one as possible are its main concerns. On the other side the Negative Binomial model; in this case the primary goal would be the interpretation of the various coefficients and their percentage of explained variance in a linear model.

Circling back to the sample size constraint related to the 4-year Olympic cycle, a Random Forest may capture complex and non-linear relationships, but in this specific context it is prone to overfitting due to specific historical anomalies. Aligning the Negative Binomial linear model provides a necessary constraint, ensuring that the relationships captured by the Random Forest remain grounded in theoretical reality.

This Dual-Model approach enhances the quality of the analysis by looking at the same goal, the prediction of the next Olympics medal table, from two different perspectives.

1.3 Thesis Structure and Research Questions

In this dissertation, the initial chapter establishes the theoretical foundation, tracing the evolution from the article of Bernard and Busse (2004) to modern machine learning applications. It explicitly identifies the "social blindness" in current literature regarding HDI and GDI indicators.

The following chapter will focus more on the construction of the longitudinal dataset, merging World Bank macro-indicators with historical Olympic performance. Moreover, the problem of missing data, ISO code synchronization, and the complexities of geopolitical noise.

Subsequently, the core of the dual-model methodology is presented in the third chapter; providing both the mathematical architecture of the Random Forest (prioritizing non-linear feature capture) and the Zero-Inflated Negative Binomial (prioritizing causal interpretability).

In the results part, the chapter evaluates the performance of the Random Forest with back testing using Tokyo 2021 and Paris 2024; it presents the feature importance rankings and coefficient analyses, culminating in the projected medal table for the 2028 Olympic Summer Games.

The final chapter synthesizes the findings to determine the cost of interpretability in sports forecasting and the role of socio-economic indicators. By evaluating the trade-off between predictive accuracy and model transparency, this study aims to provide a data-driven forecast for the 2028 Games while still giving value the socio-economic indicators that correlate with national Olympic performance.

Therefore, the primary research question guiding this study is formulated as follows: "To what extent can we accurately predict Olympic medal count while maintaining theoretical interpretability?" Connected to this inquiry, the study also tries to evaluate the performance of social indicators like HDI and GDI in predictive frameworks when compared to traditional macroeconomic variables like GDP. Additionally, examining the differences between the models in terms of trade-off and overfitting is an important question.

Chapter 2

Background and Past Literature

One of the first contributions to the analysis of Olympic data is the article of Bell (Donald (1972)) where, through a correlation-based scoring model, he was able to individuate the factors of a successful nation at the Olympics; for instance a stable and homogeneous literate population, a prosperous economy and a strong central government.

The forecasting models have become more and more advanced in the late nineties and early two-thousands; however, since 2004 (i.e. the year of publication of Bernard and Busse article), multiple changes occurred in the Olympics and in the sport industry at a general level. For example in the data that was considered by Bernard and Busse (1960-1996) one of the biggest question was "Why does China win only 6% of the medals even though it has one-fifth of the world's population?" (Bernard and Busse (2004)).

In our case this type of investigation is not valid anymore given the exponential growth of Chinese athletes since the 2000s (Hong et al. (2013)). Thus, it is evident that it was necessary to look at other more recent articles.

Li et al. (2025), for instance, use an enhanced XGBoost modeling approach that considers factors such as the recent performance of athletes or the host country's advantages, and, with the help of the grey wolf optimization algorithm, predicts the medal winners of individual countries in the 2028 Olympic Games in Los Angeles.

Zhao et al. (2025) on the other hand, address the problem by focusing on the importance of coaches using GA-BP algorithm model, combined with genetic algorithm (GA) and back-propagation neural network (BPNN). Alongside with this methodology, also Zhang et al. (2025) highlight the importance of coaches, but this study quantifies the host effect and the great coach effect separately. It proposes a formula for the host effect and evaluates the coach effect factors by segments. Finally, the study uses a MPXG framework that organically combines factor analysis, ARIMA, and XGBoost. Another paper that works with XGBoost is the one made by

Jin (2025), but in this case it was built the HEAH model framework (historical performance, engagement in sports, athlete factors, and host effect). Subsequently, in order to obtain a good result in accuracy, the hyperparameters of XGBoost were optimized using the Tree-structured Parzen Estimator (TPE) method.

One recent major advancement in Olympic forecasting that has shifted toward machine learning, most notably the two-staged Random Forest model, is the model proposed by Schlembach et al. (2022), which successfully outperformed traditional naive benchmarks for four consecutive Olympic cycles. While their model establishes groundbreaking work for predictive accuracy, the authors found that incorporating the Human Development Index (HDI) did not provide marginal improvements in prediction.

This thesis identifies this as a critical gap in the literature: while this type of social index may be redundant for absolute accuracy in a Machine Learning model (even though a similar variable such as GDP is already present), they may hold significant explanatory power. By adopting a dual-model approach, utilizing Random Forest for accuracy and Negative Binomial regression for interpretability, this research demonstrates that HDI is an important causal driver of athletic efficiency, even if in a Random Forest its feature importance is masked by macroeconomic indicators.

Parallel to the HDI, the role of Gender Development Index (GDI) will be crucial as well; in fact, due to the increasing relevance of gender studies, other authors have started to separate their datasets by gender. For example, Leeds and Leeds (2012) introduce the Total Factor Productivity (TPF), in order to capture the nation's ability to promote women at the Olympics. Alternatively, Noland and Stahler (2016) analyze sport-specific data on only female Olympic participation from 1960 to 2012, with the aim of studying the socioeconomic status of women and societal attitudes.

Finally, the article made by Lowen et al. (2016) has the goal to test the hypothesis that women's empowerment has a correlation with women's athletic success. In this specific case women's empowerment was measured using the Gender Inequality Index (GII) and athletic success by the number of participants and the number of medals won in the Olympics. The main question that the authors posed is whether the GII predicts women's Olympic success even after controlling for variables that are other predictors of Olympic success.

In this project the predictor for gender's differences will be the Gender Development Index (GDI); both indices (GII and GDI) measure disparities and inequalities between men and women in human development (health, knowledge and living standards) (UNESCO (2024)), but GDI focuses more on the ratio of female-to-male in HDI achievements (longevity, education, income) (Mauritti (2021)). Table 1 sums up the current literature and shows where this study fits in.

Table 1: Classification of Olympic Forecasting Literature and the Proposed Dual-Model Gap.

Category	Key Studies	Methodology	Core Focus
Traditional Macro-Models	Bell (1972); Bernard & Busse (2004)	OLS Regression / Correlation Analysis	GDP and Population as primary determinants.
Modern Machine Learning	Li et al. (2025); Zhang et al. (2025); Zhao et al. (2025)	XGBoost, GA-BP, HEAH Framework	Maximizing predictive accuracy; "Black Box" nature.
Socio-Economic Models	Schlembach et al. (2022); Lowen et al. (2016)	ML / Econometric Models	Inclusion of HDI, GDI, and Gender Empowerment.
Proposed Dual-Model	**Current Study (2026)**	**Random Forest + Negative Binomial**	**Bridging the accuracy-interpretability gap**.

In summary, the evolution of Olympic forecasting has reached a point where predictive power and theoretical transparency are becoming increasingly hard to combine. While the latest machine learning models achieve unprecedented accuracy, they often obscure the importance of socio-economic drivers that define athletic efficiency, such as gender equality and human development.

By adopting this dual model strategy, the social indicators will play a crucial role that was neglected by the previous literature, while still maintaining the high accuracy as a primary goal.

Chapter 3

Methodology

3.1 Data Engineering and Preprocessing

The complete dataset with all the correct variables is necessary in order to build the two models. The first main variable retrieved was GDP, whose predictive power for Olympic success is broadly recognized (Bernard and Busse (2004)). Two sources were taken in consideration and then combined for this variable; on one hand the dataset of World Bank Group that records GDP data yearly from 2016 to 2025 (World Bank (2026)), on the other hand a dataset that combined the World Bank national accounts data and OECD National Accounts data files, in order to have yearly GDP values (1960-2020) for the countries (Christy (2022)). For the common years the values of the two datasets were matching, so they were merged together.

Another indicator that is widely accepted in Olympic forecasting is the population of a country (De Bosscher et al. (2008)), thus also this one was inserted in the dataset via the UN website (United Nations (2024)), as well as the percentage of the population that is in the athletic prime, 15-45 years old (Longo et al. (2016)).

Subsequently the addition of GDI and HDI was fundamental, through the website of Herre and Arriagada (2023).

Lastly, all the Olympic related variables, such as the number of past medals won, the team sizes and the host countries, were added via the TidyTuesday package on R (Hughes (2022)).

It is also important to highlight that only the recent Olympics were taken into consideration (Atlanta 1996 onward) for two main reasons: competitiveness and geopolitical issues. In fact, historically, professional athletes were barred from the Olympics; it was not until 1986 that the International Olympic Committee (IOC) voted to allow professionals to compete, a change that officially took effect at the 1988 Games (Monsebraaten (1986)).

Regarding the geopolitical issues, establishing the 1996 Atlanta Olympics as a chronological

baseline is justified by the profound systematic realignments of the late twentieth century, most notably the dissolutions of Yugoslavia, Czechoslovakia and USSR.

For instance, if a Random Forest model is trained on historical features collected under a consolidated entity like the USSR, the subsequent dissolution of that state complicates feature attribution. Retroactively divide and reallocating historical variables to multiple successor states creates a severe data-continuity problem that can distort the model's training data.

The list of all the variables is presented and summarized in the table below:

Table 2: Summary of Variables Utilized

Category	**Variable Name**	**Description**
Identification	ISO code	Three-letter country code (ISO 3166-1 alpha-3).
	Olympic year	The specific year of the Summer Olympic Games.
Performance	Gold	Count of first-place finishes.
	Silver	Count of second-place finishes.
	Bronze	Count of third-place finishes.
	Total	Aggregated count of all medals won.
	Weighted Score	Performance metric assigning higher weights to Gold/Silver.
Lagged Features	Previous Total Medals	Total medals won in the preceding Olympic Games.
	Prev. Weighted Score	Weighted performance score from the preceding Games.
Socio-Economic	GDP	Gross Domestic Product (constant international $).
	HDI	Human Development Index.
	GDI	Gender Development Index.
	Population	Total national population.
Demographic	% Athletic Prime	Percentage of population within the 15–45 age cohort.
	Team Size	Number of athletes qualified and sent to the Games.
Contextual	Is Host	Binary indicator (1 if host nation, 0 otherwise).

A critical adjustment to this dataset is the restriction to Olympic years only, given the fact that including inter-Olympic (e.g. 2013-2015) years would introduce "noise". As a matter of fact, since Olympic medals are a discrete quadrennial outcome, treating annual GDP as a predictor for a "zero-medal" year would erroneously signal to the model that nations with an high GDP are prone to failure.

So, in order to preserve the information contained in those years without distorting the model, Lagged Variables were built as well. Macroeconomics indicators are expressed a t-1 values relative to the Olympic year, ensuring that the model predicts 2024 performance using 2023 data, thus eliminating look-ahead bias.

An additional methodological layer involved the treatment of ex post facto medals reallocations, particularly due to retroactive doping disqualifications. Thus, it was necessary to decide if using the medals awarded on the podium or the clean adjusted finalized years later.

In this case, the choice was utilize the medals awarded at the Olympic for mainly two reasons: the first one because uniformity in how data is collected is crucial and the International Olympic Committee (IOC) and anti-doping agencies can take up to a decade to retroactively strip and reassign medals. If the ex post facto count is used for London 2012, but not for Paris 2024 (many cases won't be settled by 2028) the model would employ data generated by two completely different processes. This, to a Machine Learning model, looks like noise or inconsistency. The second reason is to represent the athletic power of that moment; medals won on the day of the event reflect a country's actual athletic performance and capability at that specific moment in time. Even if an athlete is later disqualified, the fact that a country put an athlete on the podium means their sports program was operating at a certain level during that Olympic cycle. This capability signal is more valuable than legal corrections made years later to a small fraction of athletes.

A similar issue, with two different alternatives for the same data, was encountered for Team Size; the crowd-sourced dataset (e.g., Wikipedia) and verified event-level results databases (e.g., the utilized Kaggle dataset). This study exploits the latter, utilizing active competing team sizes rather than nominal delegation numbers. Several reasons justify this choice.

First of all, the datasets that include athletes who Did Not Start (DNS) or served as non-competing alternatives or bench reserves have competitors with a 0% objective probability of winning a medal on the day of the event. Thus, by selecting active Olympians the Kaggle dataset filters out irrelevant statistical noise.

Secondly, for standardization choice; the Wikipedia page for a highly developed nation may meticulously document every reserve athlete, administrative staff member, or specialized training partner all included in the "delegation" metric, whereas pages for smaller or developing nations often restrict their tallies to primary competitors. In contrast, the Kaggle dataset has a more deterministic and standardized criterion: an athlete is only logged if they appear on an official, finalized event start list.

Lastly, even though the Team Size variable is used alongside GDP in the model, there is the

risk of overfitting due to severe multicollinearity, as the "reserve player" variance simply echoes the financial capacity of the nation. By using only active athletes instead, the variable captures more uniquely specialized athletic qualification capacity.

Instead, the discourse for the team sizes of Russia and Belarus for 2024 needs a special digression; due to the geopolitical sanctions, those nations are basically absent at that Olympic. If left unadjusted, these defective entries would force the Random Forest to misinterpret a temporary political ban as a permanent structural decay in athletic capacity. For this reason, the 2024 indicators were restored to reflect a baseline capacity of 71 medals and a lagged team size of 335 athletes for Russia, and 7 medals with a lagged team size of 101 athletes for Belarus (results of 2021). This adjustment ensures that the 2028 forecast evaluates these nations on their systemic human and infrastructural capital rather than transitory political restrictions.

Another fundamental challenge in merging high-frequency athletic outcomes with low-frequency socio-economical indicators is the prevalence of missing data. Organizations such as the United Nations and World Bank operate on a 1-to-2 year delay, resulting in blank entries for recent years metric (for instance, the 2024 HDI). To prevent systematic loss when generating time-lagged variables, a Last Observation Carried Forward (LOCF) methodology was implemented. This due to the fact that indicators like HDI, GDI and Population move very slowly, thus forward filling the most recent data into the missing year ensures that the $t-1$ lag function accurately captures a nation's pre-Olympic socio-economic baseline.

Other minor adjustments involved the ISO codes; since not all datasets initially used them, any non-standard formats were converted to the correct ISO codes.

3.2 Model - Random Forest

The Random Forest is a Machine Learning method that combines multiple uncorrelated trees in its training phase and outputs the mean prediction of the individual trees (Breiman (2001)).

Through bagging and the selection of different features combination, the practice of training each decision tree in the random forest on a different subset of examples (Google Developers (2026)), this algorithm is able to mitigate the risk of overfitting (Barreñada et al. (2024)). It efficiently captures highly non-linear relationships between certain variables, such as demographic variables and human development indices, and the final athletic output.

The model was derived using the standard implementation `scikit-learn` library in Python (Pedregosa et al. (2011)).

The split between training and testing was done in correspondence with the Olympic year; in this way, the model is trained with the previous Olympic data and it is tested on the Olympic that has to predict (the model will be tested on Tokyo 2021 and Paris 2024 as well). This passage is fundamental in order to avoid any data leakage errors.

The predictive efficacy of the Random Forest can be evaluated using two complementary performance metrics: the coefficient of determination (R^2), which measures the proportion of total variance explained by the model, and the Mean Absolute Error (MAE), which provides an easily interpretable, scale-accurate metric of the average medal deviation per country after the model fit.

Another fundamental metric in the construction of the Random Forest in Python is the inclusion of `permutation importance`.

While Random Forests automatically calculate Feature Importance, this other enhanced technique estimates the contribution of each variable to a fitted model's statistical performance on a given dataset. This method is particularly useful for non-linear estimators and involves randomly shuffling the values of a single feature while observing the resulting change of the model's score. By breaking the relationship between the features and the target, it is possible to observe how much the model relies on a specific feature (Scikit-learn Developers (2026)), thanks to the resulting drop in the overall evaluation metric.

Zero-Sum Normalization

An important adjustment to the methodology of the Random Forest has to be specified.

As a matter of fact, given the fact that a standard Random Forest predicts each country as an independent unit, the unconstrained raw predictions fail to recognize that the Olympic medal pool is a finite, closed system. To adjust for this, a post-estimation Zero-Sum Normalization protocol is introduced. Let $M_{pred,i}$ represent the raw point-prediction for country i, and C represent the fixed total medal pool available at the Games (anchored to the Paris baseline of 1,044 medals). The finalized, redistributed medal forecast ($M_{adj,i}$) is calculated as:

$$M_{adj,i} = M_{pred,i} \times \left(\frac{C}{\sum_{j=1}^{N} M_{pred,j}} \right)$$

This formula ensures that the final forecasts remain structurally grounded within a realistic, finite competitive space. This scenario forms the foundation for the backtesting validations of Tokyo 2021 and Paris 2024, as well as the simulations for Los Angeles 2028.

Geopolitical Restrictions

For LA2028, however, the model requires adjustments for external constraints; indeed it is important to notice that there are geopolitical conditions regarding the status of sanctioned states that already caused issues in the prediction of Paris 2024. In fact, the solution of restoring past data was already explained at the end of 3.1, but, for the predictions of Los Angeles 2028, the theoretical framework splits into two distinct analytical scenarios:

1. **Geopolitical Normalization - Inclusion:** This scenario models a return to a standard hierarchy, with the inclusion of Russia and Belarus. The 2024 overwritten performance is then used and the model predicts the 2028 outcomes based purely on systemic human and infrastructural capital.

2. **Continued Sanctions - Exclusion:** This scenario simulates a continue ban in 2028. Even though the core Random Forest engine remains trained on the normalized and data, the sanctioned states are constrained to zero prior to prediction. The algorithm's Zero-Sum Normalization protocol is then applied, mathematically forcing the redistribution of their theoretical medal share to the other nations.

This bifurcated framework ensures that the model correctly separates political restrictions from macroeconomic athletic capacity.

3.3 Model - Negative Binomial

During the construction of this model as well several data transformations and diagnostic steps were required in order to finalize the model specification. For instance all the variables, previously cited, were temporally aligned using a $t-1$ lag relative to the Olympic target year. This strict lag structure preserves predictive validity and eliminates potential data leakage.

Secondly, the absolute scale variables, specifically population and Gross Domestic Product (GDP), were log-transformed (ln). This was necessary in order to manage extreme right-skewness an highly unequal variances (heteroscedasticity) across countries of drastically different scales.

A preliminary analysis that was carried before the actual Negative Binomial model was the Principal Component Analysis, in order to see if there are structural multivariable dependencies that could undermine the stability of a regression.

The resulting PCA biplot (Figure 1) reveals severe structural collinearity across our primary predictors; as a matter of fact almost 50% of the total dataset variance is captured by the first component, mainly made by the ln(GDP) and the Team Size. The second component, which accounts for almost 30% of the variance, runs vertically and isolates ln(Population) in opposition to the Gender Development Index (GDI). This indicates that a nation's absolute demographic scale operates independently of its institutional gender equality profile, justifying our choice to deep dive more into socio-economical factors that function outside of absolute size.

However, it is worth noticing the fact that PC1 shows that the components of the Team Size and GDP are explaining the variance in a similar way, leading to potential destabilization of the model. This was confirmed by an initial baseline Negative Binomial estimation. In fact, when running the model, the Variance Inflation Factors (VIF) for ln(GDP) and HDI exceeded critical thresholds (VIF > 20 and VIF > 13, respectively). This severe multicollinearity introduced a classical suppression effect, artificially flipping the coefficient of ln(GDP) to a statistically

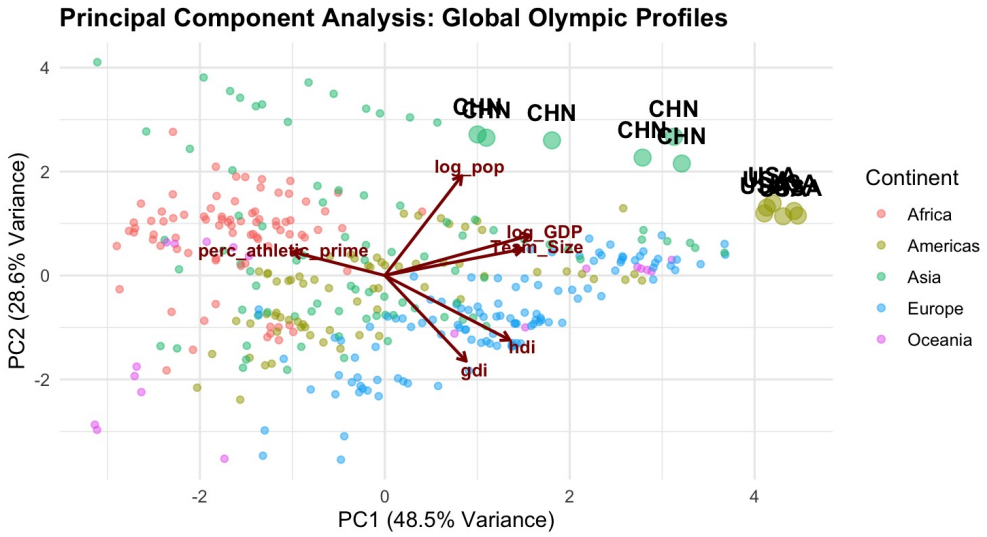


Figure 1: PCA Biplot of Macroeconomic, Demographic, and Human Development Predictors

significant negative value; a logical impossibility in Olympic macroeconomics. This is due to the fact that HDI and ln(Population), combined together, either retain raw absolute GDP or, in the case of population, possess an high correlation with GDP.

Consequently, to isolate steady and unbiased causal parameters, GDP was omitted from the regression step, and the architecture was updated to a Zero-Inflated Negative Binomial (ZINB) framework via the `glmmTMB` package.

Zero-Inflated Negative Binomial

The Negative Binomial was enhanced to a Zero-Inflated Negative Binomial (ZINB) for an important reason: standard count models, such as the Poisson or standard Negative Binomial regressions, assume that the zero outcomes are generated by the exact same stochastic process that governs the positive integers. In the context of the Olympic Games, however, zero counts are structurally heterogeneous and arise from two distinct regimes: the Structural Zeros occur when a nation has a probability of winning medals practically zero due to insufficient macroeconomic scale or athletic infrastructure. Sampling Zeros occur when a nation possess the baseline requirements to win medals, but fails in that specific Olympic due to external factors.

However, thanks to the ZINB, the model isolates this structural asymmetry and is able to acknowledge the fact that a tiny nation is in a different category respect to China or the USA.

The mathematical formulation of the mixture model is specified as follows:

$$P(Y_i = y_i) = \begin{cases} p_i + (1 - p_i) \cdot f_{\text{NB}}(0; \mu_i, \phi) & \text{if } y_i = 0 \\ (1 - p_i) \cdot f_{\text{NB}}(y_i; \mu_i, \phi) & \text{if } y_i > 0 \end{cases} \tag{3.1}$$

where $f_{\text{NB}}(y_i; \mu_i, \phi)$ represents the Negative Binomial probability mass function with conditional mean μ_i and overdispersion parameter ϕ. The structural count component is modeled via a log-link function:

$$\log(\mu_i) = \beta_0 + \beta_1 \log(\text{LagPop}_i) + \beta_2 \text{LagGDI}_i + \beta_3 \text{LagHDI}_i + \beta_4 \text{LagTeam}_i + \beta_5 \text{IsHost}_i \quad (3.2)$$

while the unobserved probability of being a structural zero, p_i, is modeled simultaneously through a logit link function:

$$\text{logit}(p_i) = \log\left(\frac{p_i}{1-p_i}\right) = \gamma_0 + \gamma_1 \log(\text{LagPop}_i) \quad (3.3)$$

Here, the dependent variable represents the `Total` medal response variable, while β and γ denote the vectors of estimated parameters for the count and zero-inflation components, respectively. By the omission of raw absolute GDP alongside the construction of this two-part mixture framework, the severe multicollinearity that destabilized the previous models was successfully resolved.

Non-Parametric Exploration via XGBoost

While the structural count component of the ZINB model assumes a strictly linear log-link relationship between the predictors and the expected medal count, macro-scale athletic infrastructure often shows diminishing marginal returns (Vagenas and Vlachokyriakou (2012)).

As a matter of fact, expanding a team from 10 to 20 athletes yields a significantly greater competitive impact than expanding another team from 230 to 240. To formally evaluate this non-linear behavior without imposing rigid parametric functional forms, an Extreme Gradient Boosting (XGBoost) model is built as an exploratory diagnostic baseline.

XGBoost is a non-parametric machine learning algorithm that optimizes an ensemble of gradient-boosted decision trees (Chen and Guestrin (2016)). It minimizes a function by sequentially fitting trees to the residuals of previous iterations, naturally capturing complex high-dimensional interactions and arbitrary non-linearities without requiring a pre-specified link function. Given the fact that the individual tree splits are black-box transformations, the marginal relationship between the infrastructure proxy (LagTeam_i) and the predicted medal output is isolated using a Partial Dependence Function:

$$\bar{f}_S(x_S) = \frac{1}{N} \sum_{i=1}^{N} \hat{f}(x_S, x_{C,i}) \quad (3.4)$$

where x_S represents the target covariate (LagTeam_i), $x_{C,i}$ denotes the vectors of complementary features for observation i, and $\hat{f}$ is the trained XGBoost prediction engine. The geometric profile of this partial dependence function indicates whether the underlying relationship is stable and linear across the feature space or whether it demonstrates structural inflection points and diminishing returns to scale.

Picewise Linear Specification

Thus, in order to maintain the causal interpretability while taking into account the non-linear relationships that will be revealed by the XGBoost, a piecewise linear regression framework is integrated into the final count component. This approach segments a continuous covariate into distinct linear intervals separated by an inflection point, or a structural "knot" (c), explicitly identified via the XGBoost partial dependence analysis. For the athletic infrastructure proxy (LagTeam_i), the variable is decomposed into a baseline slope and an auxiliary "extra" slope that activates exclusively post-knot:

$$\text{TeamBase}_i = \text{LagTeam}_i \tag{3.5}$$

$$\text{TeamExtra}_i = \max(0, \text{LagTeam}_i - c) \tag{3.6}$$

Consequently, the final structural formulation for the conditional mean count adjusts to:

$$\log(\mu_i) = \beta_0 + \beta_1 \log(\text{LagPop}_i) + \beta_2 \text{LagGDI}_i + \beta_3 \text{LagHDI}_i + \beta_4 \text{TeamBase}_i + \beta_5 \text{TeamExtra}_i + \beta_6 \text{IsHost}_i \tag{3.7}$$

Under this parameterization, β_4 captures the constant marginal effect of adding an athlete up to the threshold c, while β_5 represents the structural shift in the slope for delegations exceeding that threshold.

The last step consists in applying Cluster-Robust Standard Errors. This adjusts the p-values to account for the fact that the 8 cycles belong to the same countries, without destroying the zero-inflation architecture.

Chapter 4

Experimental Set-up and Implementation

As a consequence of all the implementations done in 3.1, the dataset is now complete and ready for the actual analysis. The finalized longitudinal panel spans eight consecutive Summer Olympic cycles, beginning with the chronological baseline of Atlanta 1996 and concluding with Paris 2024, with extra rows added at the end for the prediction of Los Angeles 2028.

In total, the dataset contains $N = 1656$ country-year observations across 17 distinct attributes. This matrix yields an important environment for macroeconomic, demographic, and performance indicators.

Table 3 outlines the definitive structural composition and missingness profile of the feature space prior to the application of the Last Observation Carried Forward (LOCF) imputation mechanism.

Table 3: Empirical Feature Profile and Pre-Imputation Missingness Rates

Variable Name	**Valid N**	**Missing (%)**	**Imputation Treatment**
`iso_code_mapped`	1,656	0.00%	None (Structural Identifier)
`year`	1,656	0.00%	None (Structural Identifier)
`Total (Medals)`	1,656	0.00%	None (Target Variable)
`Prev_Total_Medals`	1,656	0.00%	None (Inertial Lag)
`GDP_merged`	1,623	1.99%	LOCF / Historical Aggregation
`population`	1,654	0.12%	LOCF Interpolation
`perc_athletic_prime`	1,654	0.12%	LOCF Interpolation
`hdi`	1202	27.47%	Systemic Cross-Sectional LOCF
`gdi`	898	45.77%	Systemic Cross-Sectional LOCF
`Team_Size`	1,514	8.57%	Restricted to Active Competitors

The variance in missingness rates across the variables highlights the necessity of the data engineering steps described previously. While the most important economic variables, such as target medal counts, lagged historical performance, and national populations, have a near-perfect missingness percentage, instead, for the institutional indicators, is not the case.

Specifically, the Human Development Index (`hdi`) and Gender Development Index (`gdi`) lack entries for approximately 27% and 45% of the country-year matrix.

Rather than executing a deletion, the implementation of the LOCF protocol effectively takes advantage of the slow-moving structural trajectories of these indices across the four-year cycles. For variables like `Team_Size`, the 8.57% missingness rate represents historical non-participation or instances where a National Olympic Committee failed to qualify active athletes to a specific Olympic, which the modeling architecture correctly classifies as a structural zero capacity rather than an administrative omission.

4.1 Comparative Framework of the Random Forest

Regarding the characteristics of the Random Forest, in order to balance predictive performance and computational efficiency, the number of estimators (`n_estimators`) chosen was 100; preliminary testing proved that a higher number of trees increased training time without getting significantly better results. Moreover, the `random_state` was set to 42 to ensure the reproducibility of the results.

To isolate the true marginal importance of the expanded socio-economic features, two distinct modeling configurations were made:

1. **The Naive Baseline Model:** A restricted, autoregressive baseline model where the only predictor variable is `Previous Total Medals` (due to the fact that historical performance is the strongest predictor in Olympic literature (Orchard et al. (2026))), calculated via the $t-1$ lag feature.

2. **The Complete Socio-Economic Model:** An unconstrained framework integrating the full matrix of demographic shifts, human development indices, gender parity metrics, and host-nation variable, i.e the indicators shown in Table 2.

When applied to the Tokyo 2021 dataset, the Naive Model achieved a Coefficient of Determination (R^2) of 0.924 and a Mean Absolute Error (MAE) of 1.9961 medals per country. This confirms that the historical performance is a highly strong predictor in a stable geopolitical environment.

However, when applied to the Paris 2024 dataset, the Naive Model's R^2 dropped significantly to 0.768. This is due to the exclusion of the Russian Federation of that Olympic that broke historical continuity. Historically, Russia has consistently secured approximately 60 medals per Olympic; its exclusion created a significant distributional shift that the model was not trained

to anticipate. The subsequent "redistribution" of these medals among other nations creates the increased error margin.

The Complete Socio-Economic Model was executed as well.

Utilizing the Random Forest parameters defined previously (`n_estimators` = 100, `random_state` = 42), alongside the application of the Last Observation Carried Forward (LOCF) mechanism for structural indicators, the model was evaluated against the Tokyo 2021 dataset.

The Complete Model achieved an R^2 of 0.940, representing a near-perfect capture of the global variance. Furthermore, the MAE was reduced from 1.9961 to 1.3735 medals per country.

In order to determine the actual value of the increase of the R^2 (from 0.924 to 0.940), a statistical evaluation was conducted.

A Wilcoxon Signed-Rank test was prioritized over a standard Paired T-Test due to the right-skewed, non-normal distribution of Olympic medal counts. The results of the Wilcoxon test yielded a p-value of <0.01, confirming the actual importance of the inclusion of all the Socio-Economic indicators.

Moreover, to truly isolate the explanatory power of socio-economic indicators, such as GDP, population, and infrastructure, a third model was constructed by excluding the two biggest predictors (`Previous Total Medals` and `Team_Size`).

Even without the power of historical medal counts or team size, this Socio-Economic Model achieved an R^2 of 0.7737.

This result confirms that while history is the best predictor of the future, over 77% of the variance in Olympic success can still be explained purely by a nation's structural wealth, demographic scale, and investment in athletic infrastructure.

4.2 Implementation of the Negative Binomial

As explained in 3.3, the variables were log-transformed and the GDP was omitted. The initial ZINB model was fitted to the complete-case matrix, following the specifications of 3.2.

Post estimation diagnostics were conducted to verify the integrity of the model; specifically, randomized quantile residuals were simulated utilizing the DHARMa framework. This approach maps the discrete count residuals onto a continuous, standardized uniform distribution, allowing for classical visual diagnostics. However the resulting residuals versus predicted plots revealed structural misclassifications.

The quantile lines under a correctly specified model should remain perfectly flat and horizontal across the predicted values; instead here they present a non-linear curvature. This "wave" pattern in the plot indicates that the rigid log-linear link function of the baseline ZINB model was forcing a constant slope across the entire dataset, thus systematically under-predicting and over-predicting the medals.

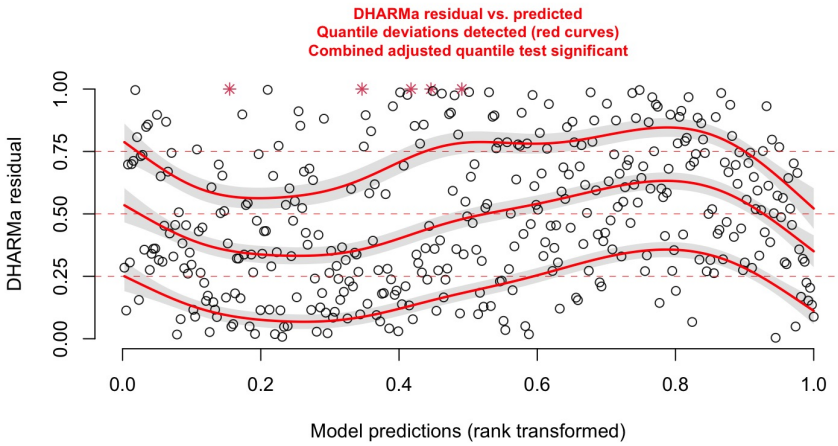


Figure 2: DHARMa residuals vs. predicted

To isolate the specific predictor creating this issue, a Generalized Additive Model (GAM) was fitted as an exploratory diagnostics. By replacing the strict linear parameters with non-parametric thin-plate regression splines (`s(.)`), the GAM framework allows the data to control for the functional shape of the marginal effects (Wood (2017)).

The output explicitly identified the athletic infrastructure proxy (`Lag_Team_i`) as the primary source of the non linearity.

In fact, a visual inspection of the smooth plots revealed a significant, non-constant trajectory: the expected medal count escalated rapidly as team sizes expanded from zero, but it transitioned into a flat plateau for larger delegations.

This plateau, confirmed by an Effective Degrees of Freedom (EDF) significantly larger than 1, confirmed that the linear parametrization of the team size was the flaw causing the curved residuals, necessitating further investigations. For this reason, the XGBoost was implemented.

The XGBoost, configured with a `count:poisson` objective function to align with the underlying distribution of the target variable, isolates complex, non-linear scaling dynamics through sequential tree-splitting. The true marginal effect of the team size variable was extracted from the XGBoost model using a Partial Dependence Plot (PDP).

The empirical curve (Figure 3) visually confirmed the hypothesized diminishing return. Below a threshold of approximately 150 athletes, the function (black line) scales aggressively, proving that team size expansion produces high marginal returns for small-to-mid-sized Olympic delegations. However, at the 150-athlete mark, the function experiences a sharp inflection and completely plateaus, demonstrating that mega-delegations achieve virtually zero marginal predictive return from raw numerical scaling. From the 200 athletes mark onward, the data points are too limited in order to make statements or interpretations about the plot.

By hybridizing a non-parametric diagnostic (XGBoost) with a rigorous parametric estimator

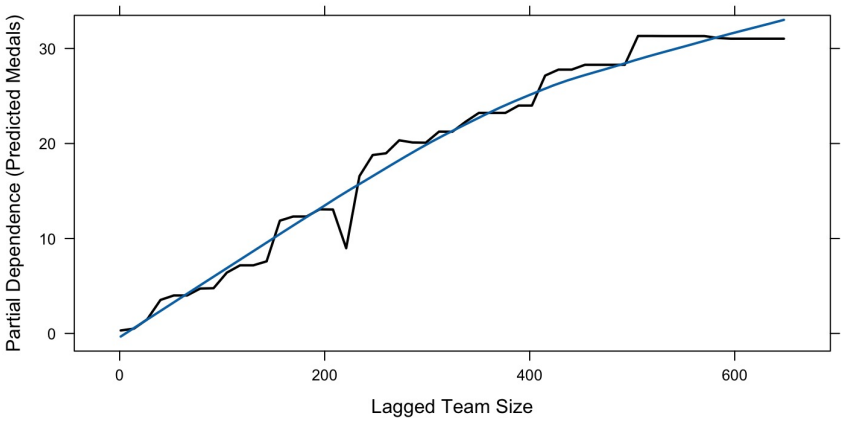


Figure 3: XGBoost Partial Dependence Plot for Lagged Team Size, revealing the structural inflection point.

(ZINB), the final Picewise ZINB specification aligns the theoretical model with the non linearity effect of the Team Size.

Finally, to ensure the validity of causal inference across longitudinal panel data, the final estimates were evaluated using Cluster-Robust Standard Errors (CR2) grouped at the country level.

This adjustment penalizes the variance-covariance matrix for the non-independence of repeated Olympic cycles.

Under this framework, 3.7 will be the baseline model for our results.

Chapter 5

Results

5.1 Backtesting Performance: Tokyo 2021 and Paris 2024

The first two models, relative to Tokyo 2021 and Paris 2024, were drafts in order to see the performance of the Random Forest before proceeding to the forecast of 2028. The raw predictions were converted from percentage shares back into absolute medal counts. This transformation utilized the definitive global medal pool from 2021 and 2024. The results are listed in the tables below.

Table 4: Tokyo 2021

$R^2 = 0.94, MAE = 0.152$

Country	Actual	Predicted
USA	113	127.1
CHN	88	103.0
RUS	71	77.6
GBR	65	69.6
JPN	58	52.0
AUS	46	29.9
ITA	40	34.6
DEU	37	48.0
NLD	36	22.3
FRA	33	49.2
CAN	24	22.4
BRA	21	27.2
KOR	20	21.1
NZL	20	16.4
HUN	20	17.2
UKR	19	14.5
ESP	17	21.2
CUB	15	12.3
POL	14	16.0
CHE	13	8.2

Table 5: Paris 2024

$R^2 = 0.80, MAE = 0.186$

Country	Actual	Predicted
USA	126	109.2
CHN	91	92.1
GBR	65	65.7
FRA	64	48.5
AUS	53	43.1
JPN	45	61.3
ITA	40	42.9
NLD	34	35.4
DEU	33	37.9
KOR	32	13.0
CAN	27	12.7
BRA	20	10.1
NZL	20	10.3
HUN	19	11.7
ESP	18	11.0
UZB	13	3.6
IRN	12	6.2
UKR	12	9.6
SWE	11	10.4
KEN	11	7.7

The "Host Effect" remains a complex variable to quantify, because the model failed to predict the number of medals for Japan in 2021 and France in 2024. A more rigorous treatment of this phenomenon will be studied subsequently.

5.2 Final Random Forest Model

The Permutation Features Importance was utilized to extract the causal hierarchy of the Complete Model.

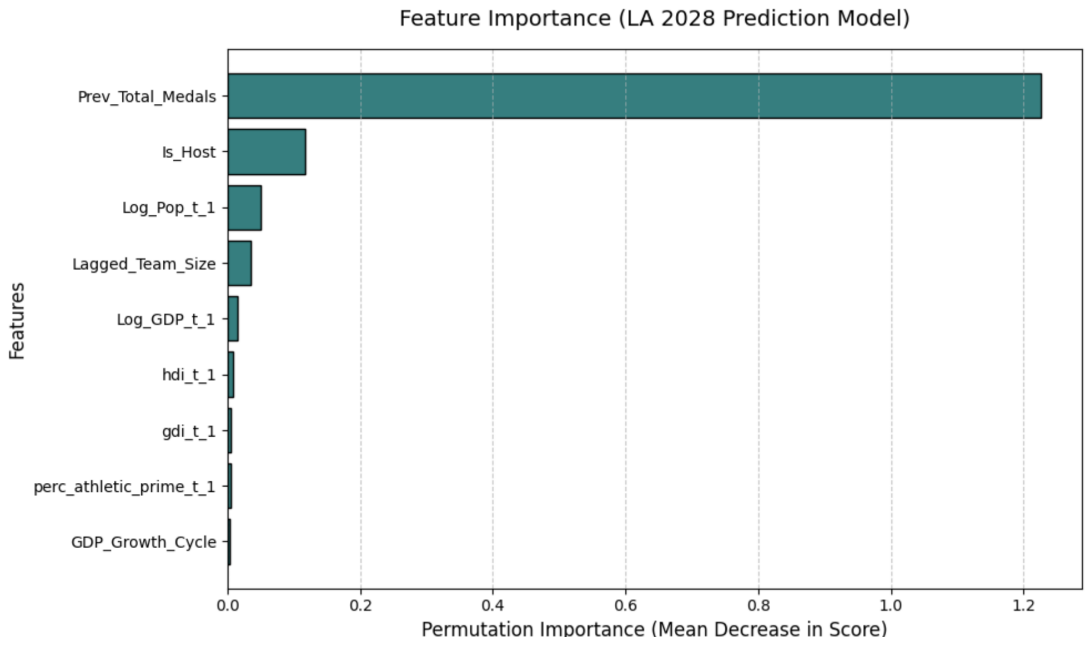


Figure 4: Feature Importance for LA 2028

As illustrated in Figure 4, `Prev_Total_Medals` emerges as the dominant predictor, confirming the statement of the existing literature (Orchard et al. (2026)). The infrastructures and the cultural capital required to produce medal-winning athletes are built over decades and rarely deteriorate over a single quadrennial cycle.

However, the predictor `Prev_Total_Medals` masks the other underlying indicators; thus, in order to show the structural foundations, `Prev_Total_Medals` was removed from the feature matrix to evaluate countries strictly on other indicators as well.

Figure 5 shows interesting findings: the Team Size predictor passes both the Hosting one and the Population one, becoming the primary structural predictor and validating its role as the direct physical proxy for athletic funding and qualification capability.

Crucially, the HDI and GDI still do not have the significant value that was expected, but a more detailed analysis of those indicators will be conducted for the results of the Negative Binomial.

The tables below present the LA 2028 forecasts, modeling the scenarios of Russia's inclusion and exclusion, respectively.

One key observation is the normalization via medal share: in order to maintain consistency

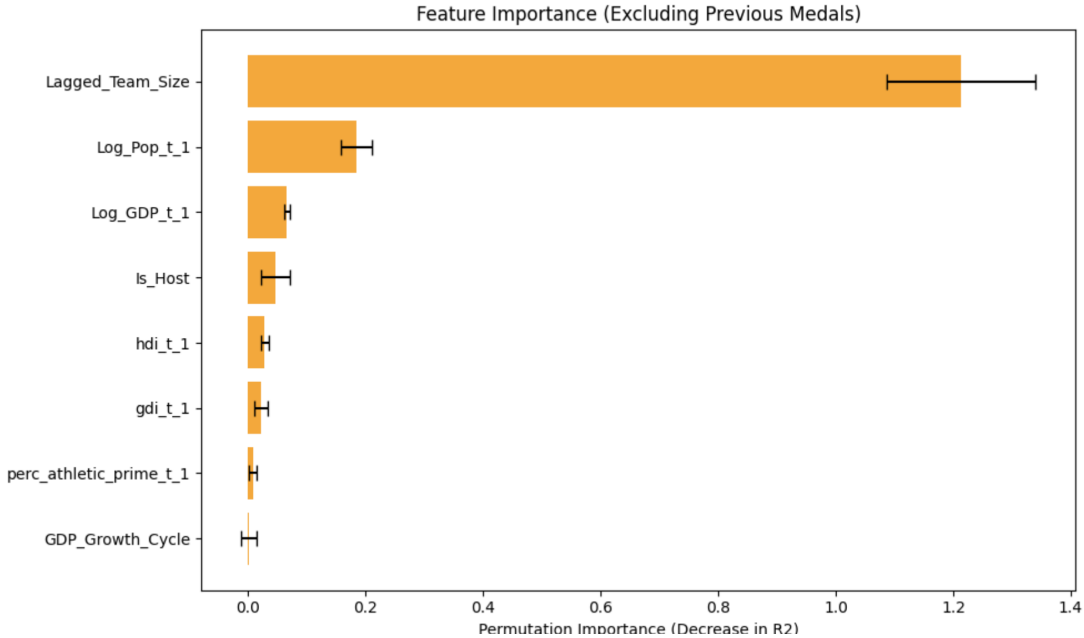


Figure 5: Feature Importance for LA 2028 without `Prev_Total_Medals`

Table 6: LA 2028 (Complete Model)

Russia Included

Country	Predicted	Share %
USA	118.0	10.12
CHN	93.7	8.04
RUS	70.6	6.06
FRA	63.9	5.48
GBR	63.6	5.45
AUS	51.3	4.40
JPN	43.4	3.72
DEU	39.6	3.40
ITA	38.9	3.34
NLD	35.1	3.01
KOR	31.4	2.69
CAN	29.4	2.53
BRA	24.0	2.06
ESP	21.8	1.87
NZL	21.2	1.82
HUN	18.1	1.55
BEL	17.6	1.51
POL	13.9	1.19
CHE	10.0	0.86
TUR	10.0	0.86

Table 7: LA 2028 (Exclusion Model)

Russia Excluded

Country	Redistributed	Share %
USA	122.51	11.73
CHN	92.30	8.84
GBR	65.35	6.26
JPN	49.89	4.78
AUS	48.01	4.60
FRA	44.63	4.27
ITA	42.02	4.02
DEU	36.20	3.47
NLD	34.65	3.32
KOR	29.11	2.79
CAN	27.26	2.61
BRA	20.59	1.97
NZL	19.99	1.91
ESP	19.43	1.86
HUN	18.77	1.80

across Olympic cycles, medal share is utilized to account for variations in the total number of sports and available medals.

Analysis on Medal Distribution

By looking at the Tables 6 and 7 there are several remarkable trends.

The United States, for instance, is projected to reach a performance ceiling in the 2028 Olympic, i.e. even an higher number of medals compared to the one predicted.

By simultaneously having an high GDP, a large population, and host-nation status, the US "maxes out" the model's predictors, likely resulting in an unprecedented medal total. Especially the host-nation variable is underestimated in both scenarios, pending the final econometric calibration in the subsequent section.

Regarding the host nation advantage, the forecasts capture the algorithmic correction for the past hosting countries. France enters the 2028 cycle with massive historical strength from Paris 2024 (64 medals and an inflated `Lagged_Team_Size`). However, under the strictly normalized Exclusion Model, France's projection regresses to 44.63 medals (a 4.27% share). This is a mathematically correction, pulling France exactly back toward its true, pre-host socio-economic baseline (having won 42 medals in Rio 2016).

Alongside this correction, Japan exhibits a similar stabilization; following their Tokyo 2021 spike (58 medals) and a mild regression in Paris (45 medals), the models project them to hold steady between 43 and 49 medals, confirming they have retained a portion of their host infrastructure but lost the direct geographic advantage.

In contrast, Great Britain, Italy and the Netherlands represent the power of structural consistency. Great Britain, in fact, operates as an example of permanent athletic development; despite London 2012 being over a decade ago, they have stabilized around 65 medals across the last three Olympics. The models perfectly mirror this reality, projecting them securely at 65.35 medals in the Exclusion scenario, maintaining their status as Europe's premier athletic power.

Italy exhibits a new baseline. After jumping from 28 medals in Rio 2016 to 40 in both Tokyo and Paris, the models confirm this was not a statistical anomaly. The Exclusion forecast projects Italy at 42.02 medals, effectively cementing their transition into a permanent top-10 powerhouse based on high human development and efficient resource allocation.

A similar reasoning applies to the Netherlands; the jump from 19 medals in Rio 2016 to 34 and 36 in Tokyo 2020 and Paris 2024 is not a coincidental spike, but the result of sustained national development and athletic investment; the model captures this underlying trend, which is why the Netherlands is projected to secure around 35 medals.

Finally, the models highlight shifting macroeconomic tendencies. Germany's projection (36.20 in the Exclusion model) accurately reflects a slow, continuous structural decline from their 2016 peak (42 medals).

Meanwhile, China solidifies its position as the primary competitor to the United States. Although the U.S. absorbs the highest absolute volume of medals in the Russia Exclusion scenario, China experiences a significant proportional gain, with its share of the total Olympic medal pool increasing from 8.04% to 8.84%. This redistribution indicates that the absence of sanctioned states disproportionately benefits the top two nations, further consolidating a bipolar concentration at the summit of the medal table.

5.3 Results of the Causal Framework

Table 8: Zero-Inflated Negative Binomial Regression Results

Predictor	**Count Model**	**Zero-Inflation Model**
(Intercept)	-6.327^{***}	17.001^{***}
	(0.029)	(4.314)
Log Lag Pop	0.146^{***}	-1.257^{***}
	(0.001)	(0.306)
Lag GDI	4.211^{***}	
	(0.020)	
Lag HDI	0.028^{*}	
	(0.010)	
Team Base	0.017^{***}	
	(0.001)	
Team Extra	-0.015^{***}	
	(0.001)	
Is Host	0.600^{*}	
	(0.004)	

Standard errors in parentheses.
$^{*}p < 0.05$, $^{**}p < 0.01$, $^{***}p < 0.001$

The estimation of the final Piecewise Zero-Inflated Negative Binomial (ZINB) yielded valid results.

The zero-inflation component confirms the foundational hypothesis regarding the importance of the population; the coefficient for (`Log_Lag_Pop`) is negative and highly significant (-1.257, $p < 0.001$), indicating that as a nation's population increases, the odds of being structurally locked in winning zero medals drastically decreases.

Within the count model, the piecewise linear spline effectively captures the diminishing return identified previously. In fact, for delegations under the 150 athletes threshold, the coefficient `Team_Base` is positive and highly significant (0.017, $p < 0.001$). By taking the exponential of this value, to obtain the Incidence Rate Ratio (IRR), the result is $exp(0.017) = 1.017$; thus for

small to mid-sized delegations every additional athlete sent to the Games increases the expected medal count by 1.7%.

However, going beyond the 150 athletes threshold becomes an insignificant advantage:

$$0.017 + \beta_5 = 0.017 + (-0.015) = 1.002 \tag{5.1}$$

Therefore, an additional athlete for that type of delegation only increases the expected medal output by a marginal 0.2%.

The most profound structural revelation of the piecewise specification is the isolation of gender equality as the primary socio-economic driver of athletic efficiency, completely surpassing broad human development metrics. This fact deserves an important digression.

HDI in the previous runs of the Negative Binomial model was meaningful factor, but the switch happened when the Team Size variable was divided.

As a matter of fact, when Team Size was forced to be a straight line, the model had unexplained variance at the top end of the distribution. Because high-HDI countries are the ones sending huge teams, HDI was falsely absorbing the credit for those medals.

By fixing the geometry of Team Size with the XGBoost derived piecewise knot, the false credit was stripped away. Once the model correctly accounts for the non linear saturation of athletic infrastructure, the GDI still survives by remaining significant and this proves that gender equality is a genuine efficiency driver.

For this reason, in the final model, the Human Development Index (`Lag_HDI`) is able to still achieve statistical significance ($p < 0.01$), however, its coefficient ($\beta_3 = 0.0277$) is practically negligible. A comparable 0.01-point improvement in HDI yields an IRR of $\exp(0.0277 \times 0.01) \approx 1.0002$, corresponding to, given the 0 to 1 scale, a negligible 0.02% increase in expected medals. Instead, the Gender Development Index (`Lag_GDI`) remains highly robust and statistically significant (4.211, $p < 0.001$).

The GDI operates also on a 0 to 1 index scale, so the IRR must be interpreted at a 0.01 increment. Thus, the transformation

$$exp(4.211 \times 0.01) = 1.043 \tag{5.2}$$

reveals that a 0.01-point improvement in a nation's gender parity yields a 4.3% multiplier to their expected Olympic medal count, holding population and delegation size constant.

This isolates the "Gender Dividend" in international sports: nations that structurally empower female athletic participation achieve a significantly higher result than nations with equivalent infrastructure but higher gender disparity.

The Hosting Indicator Digression

As noted during the evaluation of the Random Forest architecture, tree-based predictive models often struggle to accurately forecast the performance boost of hosting the games, frequently

underestimating the medals (e.g. Japan in 2021 or France in 2024). The parametric piecewise ZINB framework resolves this by isolating the "Host Effect" independently of the size of the delegation.

In fact, it is a well-documented phenomenon that host nations drastically increase their team sizes due to automatic qualification slots (Pettigrew and Reiche (2016)).

However, because this regression explicitly controls for team size via the `TeamBase` and `TeamExtra` splines, the `Is_Host` parameter does not measure the advantage of sending more athletes; rather, it strictly measures the autonomous efficiency advantage of competing domestically.

The host coefficient is positive and statistically significant (0.600, $p < 0.05$). Transforming this log-odds parameter into the IRR gives us

$$exp(0.600) = 1.822 \tag{5.3}$$

This indicates that, holding all socio-economic factors and team sizes mathematically constant, the act of hosting the Olympic Games generates an 82.2% multiplier to a nation's expected medal yield.

While an 82.2% efficiency shock may initially appear mathematically aggressive, it is heavily supported by historical baseline shifts in the modern Olympic era. For example, France experienced a 94% increase in total medals between Tokyo 2021 (33 medals) and Paris 2024 (64 medals). Similarly, China saw a 58% surge entering Beijing 2008, and Great Britain experienced a 38% jump entering London 2012.

Because the model successfully strips away the numerical advantage of the expanded host delegation, this massive 82.2% multiplier captures three underlying causal mechanisms:

1. **Targeted Financial Injection:** Host nations implement aggressive, highly funded high-performance training programs immediately upon winning the Olympic bid (typically 7 to 8 years prior to the Games), having a peak athletic cohort perfectly timed for the home Olympics.

2. **Zero Logistical Issues:** Domestic athletes benefit from a complete lack of travel fatigue, no time-zone acclimation, customized domestic nutrition, and years of exclusive practice access to the exact competition venues.

3. **Judging Bias:** In qualitatively scored disciplines (such as gymnastics, boxing and diving), there is a documented, statistically significant judging bias that historically favors the home delegation due to subconscious crowd influence (Wilson and Ramchandani (2021)), (Myers and Balmer (2012)).

To sum up, the importance and the potential effect of the host nation indicator has been proven, but it is possibly underestimated by the Tables 6 and 7. In the following chapter a

possible fix of this will be done, by integrating the two models, in order to have a final prediction as accurate as possible.

Besides this improvement, the possible enhancements of this project will be also discussed, as well as the potential critical parts.

Chapter 6

Discussion

6.1 Hosting Indicator in the SHAP Plot

The structural failure of the Random Forest in classifying categorical variables like `Is_Host` is exposed by the SHAP bee swarm plot in Figure 6.

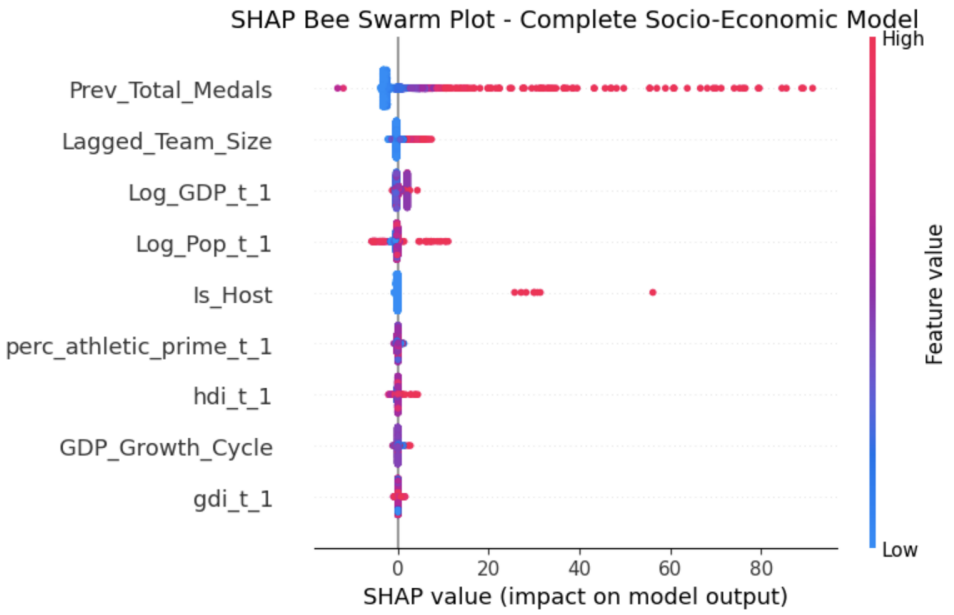


Figure 6: SHAP plot

The `Is_Host` variable, in fact, resides near the bottom of the global importance ranking. However, there is a sparse collection of red points (where `Is_Host` = 1) that extend as a right-

hand outlier vector. In this type of plot it means that they have a positive impact, but, given the fact that decision trees construct partitions by maximizing variance reduction across the entire training sample, a severely imbalanced binary flag provides little splitting utility compared to continuous indicators like `Log_GDP_t_1` or `Lagged_Team_Size`.

Consequently, the Random Forest treats the host effect as a poor predictor, even though the underlying effect is clear both in the SHAP plot as well as in the Negative Binomial analysis.

Thus, to fix this blind spot, this research introduces a Hybrid Econometric-Machine Learning Calibration; the goal is fixing the non-parametric limitations of the Random Forest by injecting the multiplicative scaling parameters of the econometric model.

6.2 The Hybrid Calibration

First, the United States' host status is suppressed in the machine learning architecture (`Is_Host = 0`). Because the Random Forest constructs its nodes based on the absolute flat medal gains of historical host nations (e.g., Japan 2021, France 2024), it applies a flat penalty of approximately 15 medals when host status is stripped, possibly underestimating the strength of a superpower. This step establishes an un-hosted, pure socio-economic baseline for the USA, i.e. 105 medals. Secondly, this baseline is multiplied by the structural log-linear multiplier for hosting 1.822, that was previously extracted in (5.3).

As a consequence the number of medals jumps to $105 \times 1.822 = 191$.

A critical methodological concern when applying this multiplicative calibration is the risk of overestimation via the double-counting.

As a matter of fact, for mid-tier sports nations (like Brazil in 2016 or Greece in 2004), hosting triggers an automatic qualification mechanism, causing their delegation sizes to increase significantly because they start from a relatively low baseline. If a model includes both `Lagged_Team_Size` and a host multiplier, it risks double-counting this infrastructure spike.

However, the United States functions as an outlier; they qualify for virtually every Olympic event on pure competitive merit, sending 592 athletes to Paris 2024 and a projected maxed-out allocation of approximately 600 athletes to Los Angeles 2028.

Because their `Lagged_Team_Size` feature is already operating at a absolute capacity, the Random Forest does not register an artificial team size explosion. Consequently, applying the Negative Binomial multiplier to the USA remains mathematically valid; it isolates pure home advantages, such as crowd psychology, the total elimination of travel fatigue, and localized officiating bias, without double-counting delegation expansion.

Nevertheless, despite the clean isolation of the home advantages, the resulting output of 191 medals exposes a limitation of applying average parametric multipliers to right-tail extremes. The multiplier of 1.822 is an aggregate coefficient calculated across all historical hosts. When a

mid-sized nation hosts, an 82% increase is highly feasible because they are climbing from a low baseline.

For a superpower, this multiplier breaks down against a physical constrain. The Olympic medal pool is a finite system of around 1000 medals. The United States cannot physically expand its medal output by over 80 medals because there are simply not enough leftovers event in the global pool that the nation does not already dominate.

This structural limit introduces a alternative perspective. Because the `Is_Host` indicator is explicitly captured within the Random Forest and mapped clearly in the high-resolution SHAP interval, the machine learning algorithm may already be capturing the true, localized, non-linear bound of the host effect completely. and recognizing that a superpower cannot break the theoretical ceiling of the pool.

Moreover, forcing a manual hybrid override introduces massive systemic fluctuation into the global forecasting model. In a closed competitive space, awarding 191 medals to a single nation forces a radical post-hoc redistribution across all remaining countries. Artificially compressing the medal shares of other elite nations (like China or Great Britain) to pay for a mathematically inflated US host multiplier risks distorting the empirical weights learned by the machine learning algorithm. While the parametric hybrid approach excels at identifying isolated causal dynamics, the uncalibrated, non-parametric bounds of the Random Forest may offer a more structurally sound reflection of a finite medal pool.

Ultimately, this alternative perspective is presented as an original hypothesis of this dissertation, offering a theoretical foundation for a possible mathematical development in future research.

6.3 Socio-Economic Drivers in the SHAP Plot

To uncover the underlying aspects of the socio-economic drivers, a truncated, high-resolution view of the SHAP values is implemented across the [-5.0, 15.0] interval. Isolating this specific window strips away the distortion of the outliers and focuses on the other macroeconomic indicators.

This visualization exposes dynamics that are obscured in the global plot:

1. **The Demographic Penalty**: Low population values (represented by the dark blue and purple points) form a pronounced left-hand tail that extends in the negative part of the plot. This visual pattern provides empirical proof of an expected demographic penalty; a small population size acts as a rigid, non-linear structural constraint that heavily drags down a nation's predictive baseline, regardless of its financial wealth.

2. **The GDP Effect**: The zoomed interval exposes an unusual vertical clustering pattern for `Log_GDP_t_1` between the +2.0 and +2.5 SHAP interval. Rather than a smooth, continuous gradient, the data points converge into a dense, vertical bar. This indicates an algorithmic

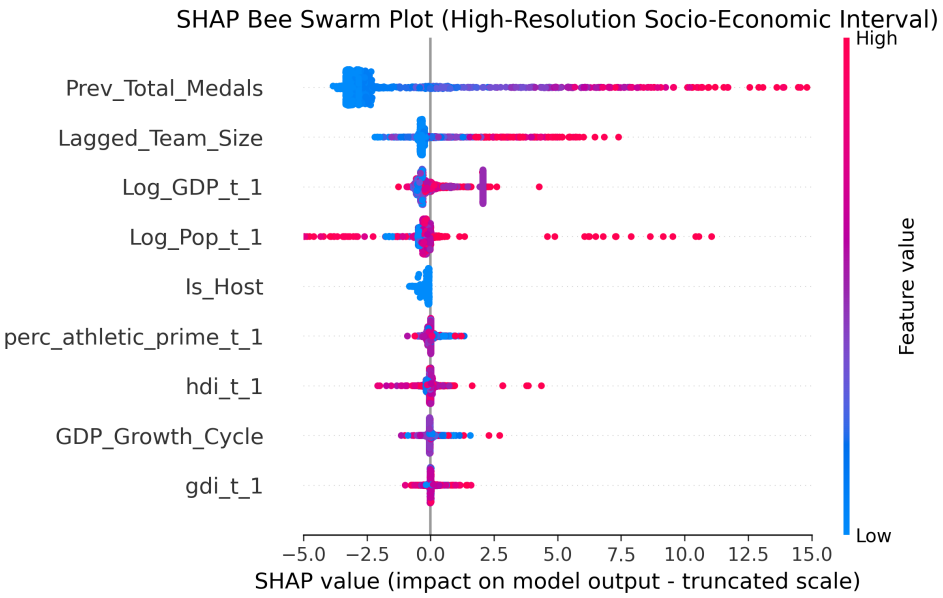


Figure 7: SHAP plot - zoomed in

threshold effect within the Random Forest architecture; in fact, once a nation's economic volume crosses a critical macroeconomic baseline, the ensemble model applies a discrete structural bump of approximately 2 to 2.5 medals.

3. **The Counter Effect of the Hosting Indicator**: In this case, the plot shows for `Is_Host` only a dense, highly concentrated stack of blue points centered directly on zero, with virtually no spread. This confirms the fact that the Random Forest fails to create a meaningful split for the vast majority of the nations, given the fact that hosting is an extremely rare event.

6.4 Limitations and Future Research Directions

While the dual-model framework successfully addresses the "social blindness" of the Random Forest by integrating the Negative Binomial model, as well as stating the issue of the hosting nation indicator, several structural limitations remain, leaving important paths for possible future expansions and optimizations.

1. **Migration Dynamics**: The current variable regarding the population act as closed pools, by assuming a static relationship between a nation's physical boundaries and its athletic talent. Modern sports, however, are deeply impacted by migrations and naturalization pathways. Future iterations should incorporate a "Migration Index" to capture both how efficiently nations use their immigrant talent as well as how they lose it with emigration.

2. **Financial Aspect:** This project relies on macroeconomic capacity via the GDP as an indicator for financial power. While highly robust, this indicator fails to isolate the actual allocation of the capital. A nation may posses a massive economy, but direct low funding towards elite sports infrastructure, coaches and allowances for the population to pay for the facilities. Another variable with those specific details included would be a more efficient and precise indicator of money spent in sports.

3. **Winter Olympic Expansions:** The dataset constructed for this thesis is restricted to the Summer Olympic Games. Expanding the dual-model framework to the Winter Olympic cycle represents a necessary test for the socio-economic variables. The Winter Olympics feature a radically different competitive scheme, where geographical, climatic, and specialized infrastructure variables (e.g., alpine access, average sub-zero degrees days, indoor ice rink density) interact with national wealth. The evaluation of the model for Winter Olympic could validate the universality of this methodology.

Chapter 7

Conclusion

This research has demonstrated the predictable nature of the Olympic Games, by engineering the national prosperity, human development and institutional efficiency.

The core contribution centers on the development of a Dual-Model Architecture. By examining together the raw predictive accuracy of a Random Forest against the structural transparency of a Zero-Inflated Negative Binomial model, this thesis has navigated the precision-interpretability trade off. The empirical findings expose the "social blindness" of historical forecasting frameworks.

While raw macroeconomics variables, like GDP and population, set the boundaries of a nation's capacity, human development and gender equity ones, like HDI and GDI, act as the quality mechanism that govern how effectively that scale is converted into podium finishes.

Ultimately, this study proves that machine learning techniques and classical econometrics do not have to be in isolation. The mathematical limitations of an ensemble learner can be systematically corrected by the structural parameters of a log-linear model.

As sports analytics and predictions shift towards black-box implementations, this project establishes a different approach, socially grounded and statistically rigorous, to macro-level sports forecasting.

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I would like to begin by expressing my gratitude to my supervisors: Arno and Harald. Arno introduced me to the intriguing field of Sports Data Science and guided this project from the beginning. His suggestions were always precise and insightful, making our meetings not only useful but genuinely interesting. We may have occasionally gotten a bit off track talking about languages, but in the end, we always managed to pull it all together!

Harald joined a bit later this project, but he took on his role exactly as I had hoped. His meticulous attention to detail pushed me to continually refine the thesis. I am especially grateful for his genuine dedication from day one, even though this topic falls slightly outside his primary area of expertise.

On a personal note, I want to thank all of my friends, you know exactly who you are!

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Finally, I want to express my gratitude to my family: my dad, mum and sister. Living so far apart has made me realize just how valuable our time together is. It might be a cliché, but distance truly does make you appreciate the value of the little things you once took for granted.

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Appendix: Additional detail

All the code for this research can be found in the following GitHub repository: link.

Appendix on Generative Artificial Intelligence use

Google Gemini was used to assist with specific parts of the research and report. Specifically, the tool was applied in the following ways:

1. Gemini was used to brainstorm research ideas, enhance the general level of the thesis and correct possible flaws, using prompts such as: "Which are other variables that could be taken in consideration in this project?", "Is there literature confirming / denying this specific statement, if so, how can I integrate it in my thesis?"

2. Gemini was also used for spelling, language and clarity checks through prompts such as: "Where can enhance or correct the clarity or grammar in this section?"

3. Gemini assisted in the coding part, specifically by helping the development of the models in Python and R, evaluating coding errors and fixing mistakes.

All the main work, including the thesis idea, data collection and analysis, the primary construction of the models and interpretation of the results, was conducted autonomously or in consultation with supervisors.

Gemini served as a supplementary tool for improving structure, code and general clarity, and all of its outputs were evaluated in a critical way.

As such, this thesis represents my own work, ideas and interpretations.